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Model Risk Management Checklist for AI/ML

Complete checklist, validation scope, and governance framework

Generated: May 11, 2026

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Table of Contents

1. Executive Summary

The Federal Reserve's SR 11-7 guidance on Model Risk Management applies to AI/ML systems in financial institutions. This comprehensive checklist ensures compliance and establishes best practices for the entire model lifecycle.

2. MRM Framework for AI/ML

Three Lines of Defense

1st Line: Model developers and owners responsible for quality **2nd Line:** Risk function provides independent review and oversight **3rd Line:** Internal audit validates governance and controls

Four Key Components

Model Development: Sound methodologies, quality processes **Validation:** Independent testing and performance assessment **Documentation:** Complete and accurate model records **Governance:** Clear policies, oversight, and accountability

3. Pre-Implementation Checklist

Model Design & Development

■ Clear business objective and success criteria defined

■ Data requirements and sources documented

■ Model architecture and algorithms justified

■ Alternative approaches considered and documented

■ Limitations and assumptions documented

■ Bias and fairness considerations addressed in design

■ Privacy and regulatory requirements assessed

■ Scalability and performance requirements established

Data Preparation & Analysis

■ Data sources and lineage documented

■ Data completeness and quality assessed

■ Data governance and ownership established

■ Data biases and representation issues analyzed

■ Feature engineering and selection justified

■ Training/validation/test splits determined

■ Data refresh and update procedures established

■ Privacy controls implemented (encryption, anonymization)

4. Development & Testing Checklist

Model Training & Validation

■ Multiple models or approaches tested and compared
■ Model hyperparameters optimized appropriately
■ Cross-validation or holdout validation performed
■ Performance metrics defined and measured
■ Accuracy/precision/recall/F1 scores acceptable
■ Model stability tested (small input perturbations)
■ Model interpretability assessed and documented
■ Feature importance or SHAP values calculated

Bias & Fairness Testing

■ Protected attributes identified (age, gender, race, etc.)
■ Model performance evaluated across protected groups
■ Disparate impact analysis performed
■ Bias mitigation strategies identified if needed
■ Fairness metrics selected and measured
■ Benchmark discrimination testing completed
■ Documentation of bias assessment results
■ Plan for ongoing bias monitoring established

Stress Testing & Robustness

■ Extreme scenario testing performed
■ Adversarial example testing completed
■ Concept drift and data drift scenarios tested
■ Model performance degradation limits identified
■ Recovery and fallback procedures established
■ Cybersecurity testing completed
■ Documentation of stress test results

5. Production Deployment Checklist

Pre-Deployment Review

■ Independent validation review completed
■ Risk assessment approved
■ Governance and escalation procedures established
■ Monitoring and alerting systems configured
■ Incident response plan documented
■ User training completed
■ Documentation package complete and accessible
■ Approval from risk and compliance functions

Deployment & Transition

■ Model deployment tested in production environment
■ Data pipelines and integrations validated
■ Performance baseline established
■ Monitoring dashboards operational
■ Escalation contacts identified and trained
■ Rollback procedures documented and tested
■ Historical comparison data captured
■ Go/no-go decision documented

6. Ongoing Monitoring Checklist

Model Performance Monitoring

■ Daily/weekly performance tracking
■ Accuracy and other metrics vs. baseline
■ Input distribution and data quality monitoring
■ Prediction distribution analysis
■ Outlier and anomaly detection
■ Error categorization and analysis
■ Performance SLA thresholds and alerts
■ Monthly management reporting

Bias & Fairness Monitoring

■ Monthly performance analysis by protected group
■ Disparate impact monitoring
■ Feedback bias analysis
■ Fairness metric tracking
■ Customer complaint correlation analysis

- Regulatory examination preparation
- Documentation of monitoring results

Risk & Incident Management

- Incident tracking and root cause analysis
- Escalation procedures tested
- Quarterly risk assessment update
- Model change documentation and approval
- Retraining effectiveness evaluation
- Regulatory change impact assessment
- Third-party model validation (if applicable)

7. Governance & Documentation

Required Documentation

Document	Content	Owner	Review Frequency
Model Charter	Purpose, scope, data, limitations	Model Owner	Annually
Data Dictionary	Features, definitions, quality	Data Steward	As needed
Model Development Report	Methodology, experiments, results	Data Scientist	Completed
Validation Report	Independent testing, results	Risk Function	Annually
Monitoring Plan	KPIs, thresholds, procedures	Risk Function	As needed
Risk Assessment	Risks, mitigations, residual risk	Risk Function	Quarterly

Governance Requirements

- Model steering committee or oversight
- Change control process for model updates
- Version control and model registry
- Clear escalation paths and decision rights
- Board/management reporting on model risks
- Regular audits by internal audit function
- Regulatory examination readiness

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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For consultation on implementation, governance, and strategic deployment of AI initiatives.